BUSINESS REQUIREMENTS DOCUMENT

E-commerce Customer Retention Intelligence

Business Owner Brief | Version 1.0 | Status: Proposed

Business request: Help us understand which customers create the most value, which customers may be becoming inactive, and where we should focus retention efforts to encourage more repeat purchases.

1. Business Background

Our e-commerce business acquires customers through different channels and sells products across multiple categories. We can report total sales and order volume, but those figures alone do not tell us whether customers return to purchase, which customer groups are valuable over time, or where we may be losing future sales opportunities.

As the business owner, I want a customer-retention view that turns transaction history into clear business insights and practical priorities. The initial solution will use RFM (Recency, Frequency, Monetary Value) analysis to group customers by purchasing behaviour.

2. Business Problem Statement

We currently lack a consistent, evidence-based view of customer purchasing behaviour. We need to identify high-value customers, repeat purchasers, newer customers with growth potential, and previously valuable customers who have not purchased recently. Without this view, retention activity may be broad and inefficient rather than targeted to the customers and groups that matter most.

3. Business Objective

- Establish a clear view of customer value and repeat-purchase behaviour.
- Identify customer groups that appear engaged, valuable, new, or potentially at risk of inactivity.
- Quantify how much revenue and how many customers are represented by each group.
- Help the business prioritise appropriate retention or re-engagement actions.
- Provide a repeatable report that management can use to review customer patterns over time.

4. Key Business Questions

Business question	What the analysis should tell us
Who are our most valuable customers?	Which customers or customer groups contribute the most qualifying sales and purchase frequently?
Are customers returning to purchase?	How many customers have made multiple valid purchases, and how does repeat purchasing vary across groups?
Who may need attention?	Which previously active or valuable customers have a long gap since their most recent valid purchase?
Where is customer value concentrated?	Which customer segments account for the largest share of customers, orders, and qualifying sales?
Where should we focus retention efforts?	Which segments are large or valuable enough to merit a targeted business action?
Is customer behaviour improving?	How do customer counts, repeat-purchase patterns, and segment composition change between comparable periods?

5. Scope of Work

In scope

- Customer-level analysis using the available customer and transaction history.
- RFM segmentation using Recency, Frequency, and Monetary Value.
- A practical set of customer segments, such as Champions, Loyal Customers, Potential Loyalists, New Customers, At Risk, and Hibernating/Lost, subject to data suitability.
- Analysis of customer counts, qualifying sales contribution, order contribution, and repeat-purchase behaviour by segment.
- Filters for analysis period and relevant customer or product dimensions where useful.
- A management-facing Power BI report with an overview, segment analysis, and customer-level drill-through or detail view where permitted.
- Business observations and prioritised recommendations based on the evidence.

Out of scope for the first release

- Automatically sending marketing messages or running campaigns.
- Predictive machine-learning churn scores or claims that a customer will definitely churn.
- Customer lifetime value forecasting unless sufficient historical data and a defensible method are available.
- Attributing changes to a campaign or channel without supporting data.
- Treating simulated product costs, prices, profit, or margin as verified actual financial results.

6. Business Definitions and Measurement Expectations

Measure	Business definition / expectation
Recency	Number of days between a customer's most recent qualifying purchase and the agreed analysis date. Fewer days generally indicates more recent activity.
Frequency	Count of distinct qualifying orders placed by a customer during the agreed analysis window.
Monetary Value	Total qualifying net sales attributed to a customer during the same analysis window, calculated using an approved business rule.
Repeat Customer	A customer with at least two distinct qualifying orders in the agreed period, unless the business approves another definition.
RFM Score	A relative score for Recency, Frequency, and Monetary Value, normally on a 1–5 scale. Higher Recency score means more recent activity; higher Frequency and Monetary scores mean greater frequency and spend.
Customer Segment	A business-readable grouping based on agreed RFM score combinations and reviewed for interpretability.
Potentially At Risk	A customer whose recency has deteriorated relative to the agreed rules and whose previous activity or value makes re-engagement worth considering. This is a signal, not proof of churn.

7. Data and Business Rules to Confirm

The analyst should document assumptions and confirm them with the business before publishing results. Do not silently choose rules when they materially affect reported customer value.

Item	Initial business direction	Validation required
Analysis window	Start with a rolling 12-month period if	Confirm the reporting date, date

	transaction coverage supports it.	boundaries, and whether comparisons use complete periods.
Qualifying order statuses	Completed orders are the initial basis for purchases.	Confirm treatment of cancelled orders, returns, refunds, and partially returned orders.
Discount	The field is understood to store a decimal rate (e.g., 0.10 = 10%) applied to the whole order.	Verify actual field behaviour and whether discounts are repeated on every line of a multi-line order.
Monetary value	Use qualifying net sales, not cost or simulated profit.	Confirm the actual selling-price source, discount calculation, and any shipping, tax, refund, or currency rules.
Order grain	Frequency must count distinct orders, not transaction rows.	Confirm whether one OrderID can appear on multiple product rows.
RFM scoring	Use relative scoring across eligible customers as a starting approach.	Review score bands and segment rules with the business for meaningful outcomes.
Customer identity	Use the agreed stable customer key.	Check missing IDs, duplicate customer records, and customers with merged identities.

8. Expected Report Experience

Report area	Business purpose
Customer Retention Overview	Show total customers, repeat-customer rate, qualifying customer sales, and segment distribution at a glance.
RFM Segment Analysis	Compare segment size, sales contribution, average recency, purchase frequency, and monetary value.
Customer Detail / Drill-through	Allow an authorised user to inspect a customer's RFM scores and relevant purchase summary.
Period and Segment Filters	Allow management to review a consistent analysis window and compare relevant customer groups.

The report should prioritise readable business explanations over visual complexity. Every chart should answer a defined business question, and segment names should be accompanied by clear definitions.

9. Expected Deliverables

- A documented and validated customer-retention dataset or semantic model.
- An agreed RFM scoring and customer-segmentation methodology.
- A Power BI report that answers the business questions in this document.
- A concise findings summary identifying important customer groups and retention opportunities.
- A data-quality and assumptions note, including known limitations.
- A short handover explaining how to interpret the report and how often it should be refreshed.

10. Success Criteria / Acceptance Criteria

- Customer counts reconcile to the approved customer population and filters.
- Frequency counts distinct qualifying orders and does not inflate when an order contains multiple product rows.
- Monetary value reconciles to the approved sales definition and excludes non-qualifying orders as agreed.
- Every customer assigned to a segment has a traceable RFM score and documented segment rule.
- Management can identify the largest and most valuable segments and the segments that may warrant attention.
- The report makes no unsupported claims that inactivity proves churn or that an observed pattern was caused by a particular action.
- Data limitations, simulated values, and unresolved assumptions are clearly disclosed.

11. Roles and Responsibilities

Role	Responsibility
Business Owner	Defines the business problem, approves definitions and prioritisation, reviews findings, and decides what action to take.
Data Analyst	Profiles data, validates quality, translates business rules into measures, builds the analysis/report, documents assumptions, and presents evidence-based findings.
Data / System Owner (if applicable)	Clarifies source-field meaning, status logic, customer identifiers, refresh frequency, and access constraints.

12. Risks and Limitations

- The available dataset may not include refunds, partial returns, or actual net transaction amounts.
- A synthetic or simulated price/profit model cannot be presented as actual business financial performance.
- RFM describes past purchasing behaviour; it does not independently explain why customers behaved that way.
- A 12-month window may be unsuitable for categories with naturally infrequent purchases; segment rules may need category-aware review.
- Small customer groups, missing dates, or duplicate identifiers may distort scores and should be reported.

13. Business Owner's Closing Request

Please assess the available data, propose a defensible RFM methodology, and deliver a report that helps us decide which customer groups deserve attention and why. Where the data cannot support a conclusion, make the limitation visible rather than filling the gap with an assumption.

End of Business Requirements Document